A DOMINANT-NOISE DISCRIMINATION SYSTEM FOR IMAGES CORRUPTED BY CONTENT-INDEPENDENT NOISES WITHOUT A PRIORI REFERENCES

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ABSTRACT

A dominant-noise identification system used on images corrupted by content-independent noises without a priori information has been developed. As representative examples of content-independent noises additive white Gaussian noise (AWGN) and random impulse noise (RIN) are used. The system uses two noise detectors to estimate the intensity component of either AWGN or RIN in an image. For this purpose an impulse noise detector was developed based on directional edges [1] and a Gaussian noise intensity detector developed in [2] was implemented. Then in the second phase, lookup tables that are generated beforehand using the structural similarity index (SSIM) proposed in [3] are used to convert the detected noise intensities onto a common scale to compare the distortion caused by different noise types and identify which one is dominant. Simulations were conducted on over 300 images and results show that for images corrupted with moderate noise intensities the system's overall success rate for identifying the dominant noise is 7 out of 10 times.

***Index Terms*—** Blind-IQM, noise-type determination

1. INTRODUCTION

The demand for streaming video is constantly increasing as new multimedia applications are coming out every day. Most of these multimedia applications employ some type of display combining live video data, streaming text and user input. To maintain a high QoS while keeping up with real-time data an efficient image noise filter that can handle a wide variety of noises will become necessary. A lot of research has already been done by every major developer to design excellent noise filters. However, little thought is ever put into which type of noise filter is needed for an image corrupted by various types of noises or for a noise with properties that change. If the most appropriate filter is used, the image quality will improve. If the wrong noise filter is used, the resulting image is usually deteriorated in some way. Therefore to design an "intelligent" noise filter that can analyze an image and automatically reconfigure itself depending on the distortion, two components are needed. The first is a library of third-party noise filters that can be purchased as independent IP-blocks. The second is a component that can analyze an image and automatically choose the most appropriate noise filter. Determining this in the case when only one type of noise exists already presents certain difficulties. For the case where multiple types of noises exist in an image, the lower intensity noises that are barely noticeable or cause only minor distortions are less of a concern. The specific type of noise that causes the greatest distortion is the one that needs to be filtered out.

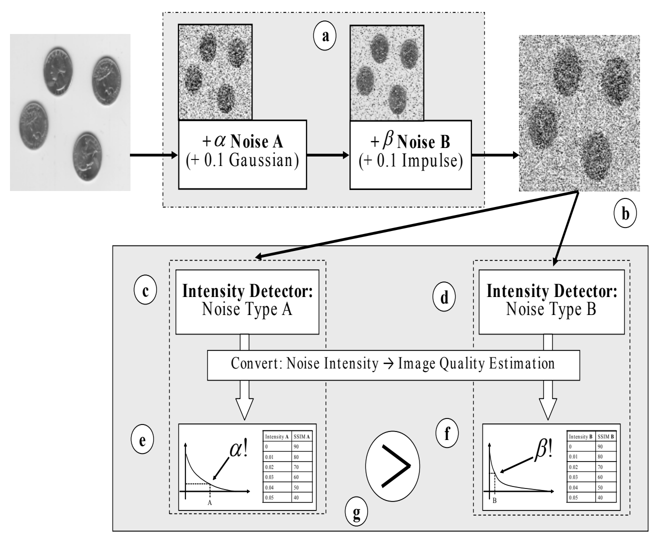

Figure 1. Automatic dominant noise-type determination

Humans can identify the dominant type of noise in an image without much trouble. However, for a computer this still remains a difficult task. The way humans do this is by comparing a distorted image to some imaginary "idealized" image recreated in the brain. However, this idealized recreation is simply the cumulative past experience of what a clear and undistorted image should look like. Thus, humans compare an image to a cumulated "database" of past experience and focus on the cause for the largest discrepancy between the distorted image and an idealized imaginary image while at the same time ignoring the minor distortions that are perhaps visible but less important.

For a computer to identify the dominant type of noise in an image, as outlined in Figure 1. first the intensity of each type of distortion must be detected and quantified (Fig. 1-c and Fig. 1-d). Then, to identify only the type of distortion that is important while ignoring the noise that cause only minor distortions, the different types of noises must be associated to each other on an equivalence scale (Fig. 1-e and Fig. 1-f) so that intensities of different noise types can be compared with each other and consequently put in order of importance (Fig. 1-g). Finally, with this knowledge, a third-party noise filter best suited to which ever noise type is dominant can be applied to the image to remove only that type of distortion. In this way the image need only be filtered once.

Both [3] and [4] give fairly good descriptions of many terms and concepts in image quality assessment (IQA). There are two types of image quality measure (IQM), those that attempt to emulate the human visual system (HVS) and try to incorporate aspects such as color differentiation, contrast sensitivity and pattern masking effect [5], or those that extract and analyze certain specific features, distortions or characteristics from an image such as edges, contours, or block artifacts [6]. The former, known as "vision modeling" systems usually try to accommodate every type of distortion at once whereas the later, called "Engineering approach", only look at one type of distortion at a time yet rarely do any of the studies done so far assume that multiple types of distortions can coexist in the image at the same time. The work done in [7] is certainly promising where they use to exploit Natural Scene Statistics in order to classify "distortion-specific signatures" however it is yet to be shown how it would respond to an image that has been corrupted by multiple types of noise at the same time. In [8] they determine the parameters of AWGN and Impulse noise present in an image using a fractal Brownian model. However, they do not go one step further and determine whether one type of noise is more disruptive to an image than the other.

Consequently, a system that can identify the dominant-noise in an image has been developed in the present work. Additive white Gaussian noise and random impulse noise were selected as representative examples of content-independent noises for the purposes of this study. The system uses noise intensity detectors to detect to quantify the component noise intensity of either noise type independently. Then, regression tables that have been generated beforehand with the use of the structural similarity index presented in [3] are used to associate the detected noise intensities to a common scale for comparison of the distortion caused by each noise type. Section 2 will discuss the noise intensity detectors and section 3 explains how the regression tables are generated using the SSIM. The Results are discussed in section 4 and concluded in section 5

2. NOISE INTENSITY DETECTORS

The first step to determine the dominant noise in an image is to detect the individual component intensities of each type of noise from the image. Each type of noise creates specific artifacts in an image, the task is to identify these artifacts and quantify them while compensating for the effect of different overlapping noise. Consequently, how well each component noise intensity can be isolated depends on how well a detector can isolate the specific features of one type of noise without being perturbed by the features created by any other type of noise in the image. The first noise detector estimates the intensity of additive white Gaussian noise in an image, and was implemented using the algorithm developed in [2]. The second noise detector quantifies the intensity of random impulse noise in an image and is described in [1]

2.1. Additive White Gaussian Noise Detector

Additive white Gaussian noise (AWGN) is an image-independent distortion that affects every pixel of an image. The intensity of the noise has a mean and a variance that determines the probability that a given pixel be corrupted and by how much the new value is different from its original value. To estimate the intensity of AWGN in an image the algorithm used was first described in [2] and is well described in both [6] and [9]. The algorithm estimates the global variance of an image by using the local variances of smaller smooth areas of an image. First the image is filtered by a cascade of 1-D filters, in both the vertical then in the horizontal direction. Then the local variance of every 3x3 window is computed. After which a histogram is created from the square root of local variances. The algorithm overestimates the variance for the pixels of higher numerical values and is compensated by implementing a fade-out function to the histogram using a cutoff function with a threshold $\beta = 2.12$ as was done in [2]. In most cases 2 iterations of applying the fade-out function was enough to diminish the "fat-tail" effect. Figure 2. shows the accuracy of the AWGN detection algorithm. If a perfect response were detected, the curve would follow the x=y line but due to limitations of the algorithm the estimate becomes less accurate at higher noise intensities as there are fewer and fewer smooth areas in the image. The accuracy of the detection algorithm could have been enhanced by introducing a compensating factor to correct this difference and shift the arc closer to a 45 degree angle, however the response is faithful enough at low noise intensities and it was thought that correcting the estimated intensity at this stage would have a more severe impact at higher noise intensities. Therefore no compensation factor was added. The intensity of AWGN noise is measured by the variance of the Gaussian function applied to every pixel.

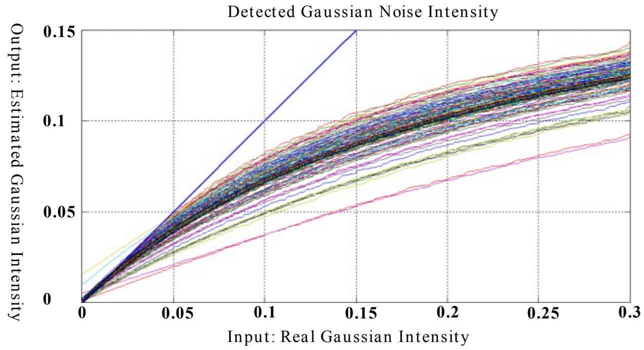

Figure 2. AWGN detection algorithm accuracy. X-axis: Gaussian intensity added to an image, Y-axis detected intensity

2.2. Random Impulse Noise Detector

Random Impulse Noise (RIN) is a content-independent distortion that affects only a fraction of the total number of pixels in an image. A number of pixels in an image are chosen at random and then each of those pixels is then assigned a new random number between the ranges of possible values. To estimate the intensity of RIN the algorithm from [1] was used.

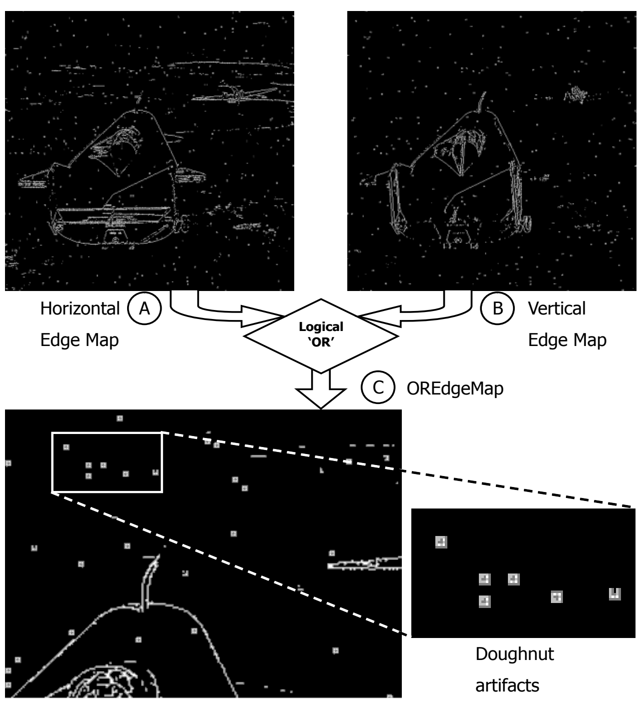

Figure 3. A) Horizontal edge map; B) Vertical edge map; C) Logical 'OR' using Horizontal and Vertical edge maps as inputs depicts 'doughnut' shapes where random impulse noise is present

The algorithm consists of passing a Prewitt horizontal edge detector over the image (Figure 3-A), and do the same with a Prewitt vertical edge detector (Figure 3-B). Then effect a logical "OR" operation between the two edge maps (Figure 3-C) and do this at every threshold value of the edge detector (0 to 255). The result are 256 edge maps, one for each threshold value, where isolated pixels that are distinct from its surroundings will display a kind of 'donut-like' artifact surrounding it (Figure 3-C). To locate these donut-artifacts a series of filters with various kernels are swept over each of the 256 edge maps to pinpoint each of these pixel locations. When ever a filter identifies a pixel, it is flagged and the results tabulated. The more often a particular pixel is flagged, the more likely it has been corrupted. Only the pixels that have been flagged more than 5 times are tabulated and the proportion of pixels identified is the intensity of pixels corrupted by Random Impulse Noise. Figure 4. shows the accuracy of the algorithm to identify RIN of 5 different images. The intensity of impulse noise is measured in "% pels", which indicate the % of pixels that have been corrupted by impulse noise.

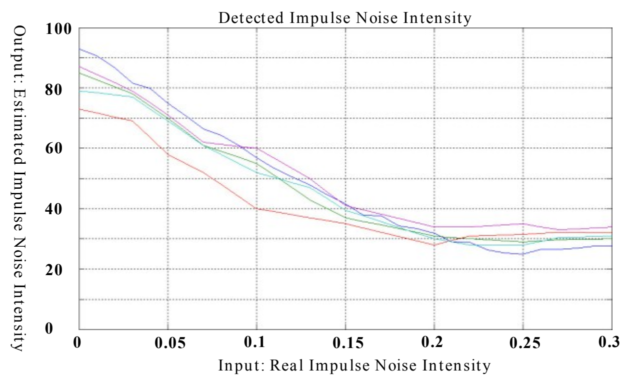

Figure 4. RIN detection algorithm accuracy; X-axis impulse noise intensity added to an image, Y-axis, the percentage of corrupted pixels which have been correctly identified as being noise

2.3. Cross-contamination

Each type of noise distorts an image in a specific way and the detector's responsibility is to identify one specific type of distortion and quantify it. However the distortions of different noise types often resemble each other and therefore noise detectors can incorrectly identify one type of distortion as being noise of another type. Also, the impact that noises have on each other has a lot to do with the order in which the noises are applied to the image. However in the case of Gaussian and Impulse noise, since both noises are content-independent the order does not have any impact.

Impact of RIN on AWGN detection

Figure 5. shows the effect that adding impulse noise has on the accuracy of the Gaussian noise detection algorithm. As an increasing density of impulse noise is added, the curve that the detection algorithm follows does not change as the impulse noise is falsely interpreted as a part of the Gaussian noise. This is due to the first step in the Gaussian noise estimation algorithm where a 1-D filter is applied to the image in essence "suppressing" the image and bringing

out many localized artifacts. Therefore, while adding extra impulse noise does affect the estimation of each local variance, fading-out the histogram has a regulating effect. This misinterpretation has a linear relation to the intensity of the impulse noise and is similar to an offset value added to the Gaussian noise intensity. To overcome this offset a simple linear compensating factor that was related to the impulse noise detector's value was added to the AWGN intensity when the two are being compared.

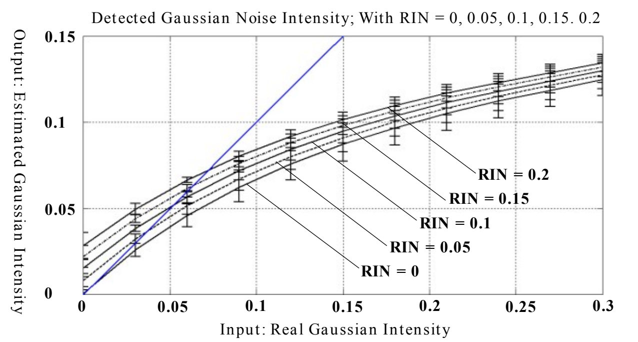

Figure 5. Impact of RIN on AWGNoise detection algorithm; for impulse noise intensities [0, 0.05, 0.1, 0.15, 0.2]

Impact of AWGN on RIN detection

Figure 6. shows the opposite example of the impact of Gaussian noise on the Random Impulse Noise detector. In this case the effect is unpredictable and non-linear when the Gaussian noise intensity is greater than 0.1 in variance and if the impulse noise is greater than 0.2% of corrupted pixel. But at these noise intensity levels images are no longer within the normal viewing range. At those levels or greater there is so much distortion in the image that it becomes difficult even for a human to determine which one causes more damage.

3. REGRESSION TABLE GENERATION

In order to determine which noise type has the strongest detrimental effect on an image they must be compared on an "Equivalent Noise Intensity" scale. This equivalence scale is generated using the Structural Similarity Index (SSIM) [3] used as a baseline for comparison. A table is generated for each noise type and converts the detected noise intensity to an equivalent modeled SSIM score generated from a database of 300 images and subjected to a wide range of noise intensities.

3.1. Table Generation

The main reasons for choosing the SSIM is that it has proven to be far more reliable than the PSNR and secondly because it is easier to use than any other Full-Reference image quality metric currently available. Using a database

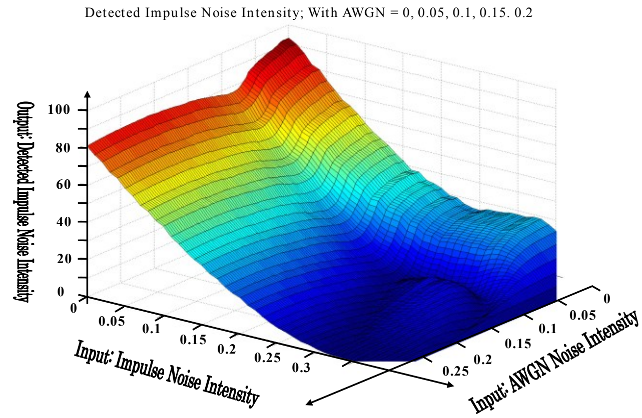

Figure 6. Impact of AWGN on RINoise detection algorithm; for Gaussian noise intensities [0, 0.05, 0.1, 0.15, 0.2]

of over 300 images gathered from various image processing databases [10][11], the images were taken one by one, and corrupted with a gradually increasing intensity of noise intensity and the SSIM score was computed and recorded at that noise intensity increment. This was done in turn first for additive white Gaussian noise (Figure 7.) for an AWGN variance intensity ranging from [0 – 0.3], and then the same was done for Random Impulse Noise (Figure 8.) for an RIN noise intensity ranging from [0 – 0.3] % of corrupted pixels. The process was repeated on the database of 300 images. Then, the mean value from all of the results of every image of one noise type is computed for both RIN and AWGN. The mean value of each Gaussian and Impulse noise represents the amount of distortion in an image due to one type of noise at a given noise intensity. The graphs were stored in table form for easy lookup during test phase. It's worth noting that images gave similar responses for both Gaussian and Impulse noise. Images that gave a response above that of the mean for impulse noise also had the same behavior for Gaussian noise. This was not the case when the graph was also plotted in another experiment using blurring noise or other data-dependant noise types.

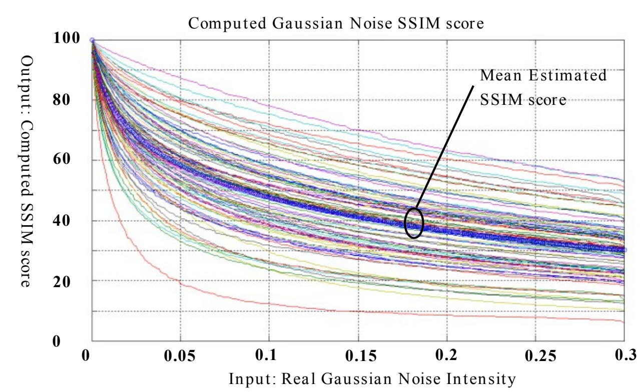

Figure 7. progression of SSIM value (y-axis) with respect to Gaussian noise intensity (x-axis) and mean value; using 300 images

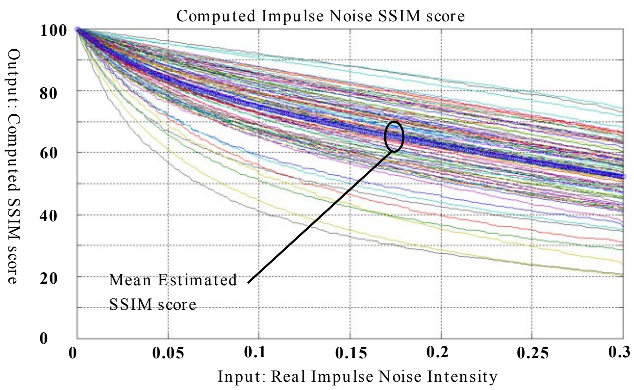

Figure 8. progression of SSIM value (y-axis) with respect to Impulse noise intensity (x-axis) and mean value; using 300 images

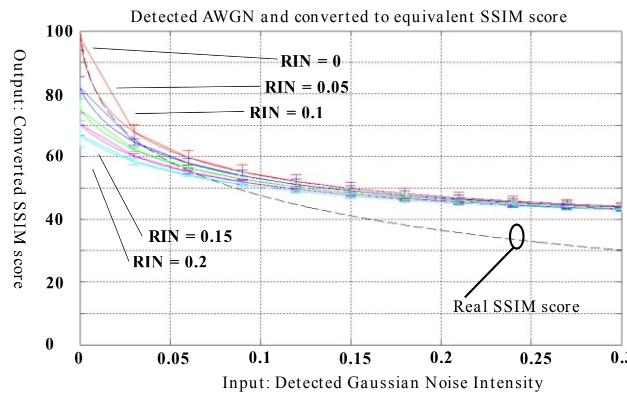

Figure 9. AWGN detection and intensity converted back through regression tables to find the estimated equivalent SSIM distortion; for RIN = [0, 0.05, 0.1, 0.15, 0.2]. The dashed-line is the Real SSIM score of the distorted image

3.2. Noise Intensity Comparing and Threshold

Section 2 discussed noise detectors for each type of noise to extract the intensity of each specific type of noise from an image. In section 3 equivalent distortion tables were generated for each noise type that represents the distortion due to one type of noise at that specific noise intensity. These tables convert the detected noise intensity onto a common scale so that they can be compared with each other. To compare the distortions due to each individual noise type, first the image is analyzed with a noise intensity detector for each type of noise to extract the intensity of each separate type of noise from an image such as the Gaussian noise detector from section 2.1. Then the result is converted to an estimated "equivalent" SSIM score using a table derived from Figure 7. recorded and put aside. The same is done for impulse noise using the noise detector from section 2.2 and a table derived from Figure 8.

Figure 9. shows the detected Gaussian noise intensity once it has been converted to its estimated equivalent SSIM score using its regression table. Each stratified curve is the detected and converted Gaussian noise intensity when impulse noise is also added to the image prior to detecting the Gaussian noise intensity (RIN = [0, 0.05, 0.1, 0.15, 0.2]). The dashed line is the real SSIM score of the distorted image that should have ideally been detected and estimated through the regression tables. A linear compensation factor was then added to the estimated Gaussian SSIM score that is proportional to the detected impulse noise intensity. No compensation factor was added to the estimated impulse noise distortion due to the unpredictable behavior of the impulse noise detector when Gaussian noise is also applied to the image as seen in Figure 6. The two estimated equivalent SSIM scores are compared to each other and the noise type with the lowest equivalent SSIM score is the type of noise that causes the most distortion in the image. Figure 10. is an example of this comparison where the impulse noise intensity is fixed at RIN = 0.1 and the Gaussian noise ranges from [0 – 0.3].

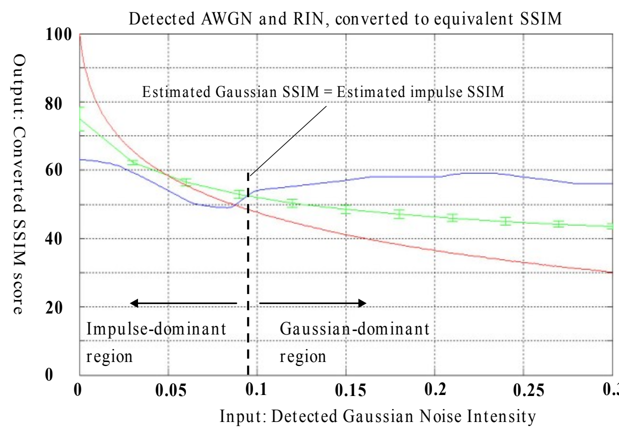

Figure 10. Green: Detected Gaussian noise (for Gaussian intensity [0-0.3]) and converted into estimated equivalent distortion. Blue: Detected Impulse noise (for Impulse intensity RIN = 0.1) and converted into estimated equivalent distortion. Red: Real SSIM score of distorted image. Left of cross point is impulse-dominant, right of cross point is Gaussian-dominant

The Green line is the detected Gaussian noise intensity converted into its estimated equivalent SSIM score. The blue line is the detected impulse noise intensity converted into its estimated equivalent SSIM score. Since the impulse noise intensity is fixed the estimated score should remain constant however the impact of the Gaussian noise can be seen. Where the Blue and Green line cross is where both estimated equivalent SSIM scores are approximately equal and at this point the distortion due to the Gaussian noise is approximately equal to the distortion due to Impulse noise. Left of the cross-point is where the estimated SSIM score of the Impulse noise is lowest and therefore the distortion of impulse noise is highest. This case is considered to be Impulse-Dominant. To the right of the cross-point is where the Gaussian noise's estimated equivalent SSIM score is lowest and consequently the distortion due to Gaussian noise is highest. In this case the image would be Gaussian-Dominant.

4. RESULTS

To test the accuracy of the system the following experiment was set up: An image was chosen at random and an intensity alpha of Gaussian noise was added to the image (Fig. 11-b). Then the SSIM score of this distorted image was computed and recorded (Fig. 11-c). Then the same clean image was corrupted with an intensity beta of impulse noise (Fig. 11-e) and the SSIM score of this image was computed and recorded (Fig. 11-f). Then an intensity beta of impulse noise is added to the image that had been corrupted of Gaussian noise (Fig. 11-d). Next the dominant-noise detection system is applied, first detecting each noise type (Fig. 11-g) and then the detected intensities are converted into their estimated equivalent SSIM scores (Fig. 11-h) the results are then compared and the noise type with the lowest estimated equivalent SSIM score is determined to be dominant. Finally this result is compared to the REAL SSIM scores determined at the beginning (Fig. 11-c and 11-f) to verify if the noise type that was deemed dominant is correct. Figure 12. shows the accuracy of the system.

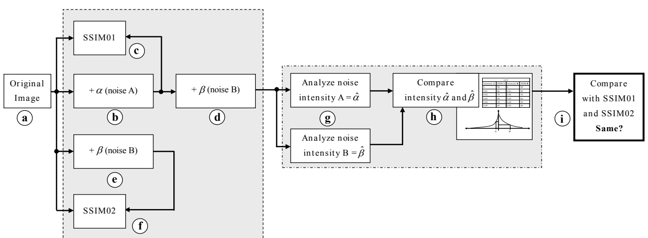

Figure 11. Test image generation and automatic noise-type determination

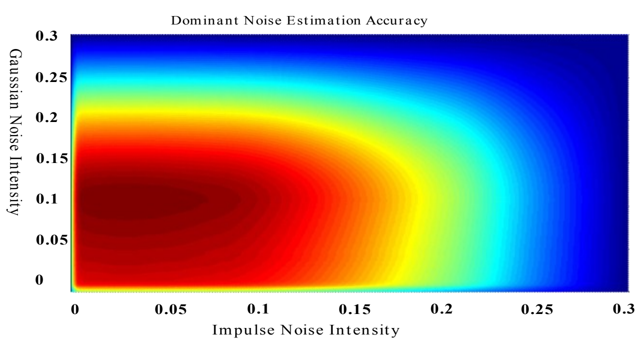

Figure 12. accuracy of the dominant noise-type detection system

The results from the experiment were tabulated, if the correct dominant-noise type was detected, a score of +1 was recorded for that noise-intensity. If the detected dominant-noise was incorrectly identified, a score of -1 was recorded for that noise intensity. In this way, if the system randomly detected the result, the negative scores would remove any random positives. The red region is greater than zero, means the system was correct more times than it was incorrect. The blue region is below zero, means the system was incorrect more times than correct. The yellow line approximately crosses the zero-point.

5. CONCLUSION

A method to analyze a non-referenced image that has been corrupted by a random intensity of either AWGN or RIN or both has been developed. The system uses two detectors to quantify the intensity of both the impulse noise and the Gaussian noise. Then, using a reference scale generated with the help of the Structural Similarity Index, the system assigns an estimated distortion level to the intensity detected by each noise detector and then determines which of the two has the greater negative impact on the image. For these specific noise intensity detectors, simulations results show that for images corrupted with moderate noise intensities the system's overall success rate for identifying the dominant noise is 7 out of 10 times. The accuracy of this method depends on the selectivity of the noise intensity detectors to discriminate between one type of noise and another. As these noise intensity detectors improve, so would the overall dependability of this type of system. The second dependency is the "equivalent noise intensity tables" generated using the SSIM. As new full-reference image quality metrics come out that better model normal human perception, so then will the regression tables follow a more human-like response.

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